

JEE MAIN REVISION MASTER: STATES OF MATTER 2026 0 QUESTIONS 0% WEIGHTAGE

Subject: Chemistry | Target Scope: JEE Main Previous Years | Empirical Base: 8 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
High	Kcet States Of Matter - Core Application	8 PYQs	General Concept

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
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3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 Kcet States Of Matter - Core Application (High Yield — 8 PYQs)

Core Principle: Key problem pattern in Kcet States Of Matter.

■ **Caution / JEE Pitfall:** Watch for algebraic and sign errors.

■ **Speed Trick:** Verify dimensions and boundary conditions.

Representative Source PYQs:

- [2024 Shift] Q#1 (MCQ): "Identify the wrong relation for real gases :..." [Ans: N/A]
- [2022 Shift] Q#2 (MCQ): "Which property of CO₂ makes it biologically and geo-chemically important?..." [Ans: N/A]
- [2022 Shift] Q#3 (MCQ): "The volume of 2.8g of CO at 27°C and 0.821 atm pressure is (R=0.08210L. atm Kmol⁻¹)..." [Ans: N/A]

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: KCET 2022 (Q#2)

Which property of CO₂ makes it biologically and geo-chemically important?

- (A) Its colourless and odourless nature
- (B) Its low solubility in water
- (C) Its high compressibility
- (D) Its acidic nature

Correct Answer: None

Solution Analysis: The property of low solubility of carbon dioxide in water makes it biologically and geo-chemically important. Carbon dioxide on reaction with water form carbonic acid, which dissociates to give HCO₃⁻ ions. H₂CO₃/HCO₃⁻ buffer system helps to maintain pH of blood between 7.26-7.42 .